

Bangladesh Studies
Agriculture of Bangladesh
Article 03

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1 Agriculture of Bangladesh

Bangladesh is primarily an agrarian economy. Agriculture is the single largest producing sector of economy since it comprises about 23% of the country's GDP and employing around

62% of the total labor force. The performance of this sector has an overwhelming impact on major macroeconomic objectives like employment generation, poverty alleviation, human resources development and food security. Meeting the nation's food requirements remains the key-objective of the government and in recent years there has been substantial increase in grain production. However, due to calamities like flood, loss of food and cash crops is a recurring phenomenon which disrupts the continuing progress of the entire economy.

Agricultural holdings in Bangladesh are generally small. Through Cooperatives the use of modern machinery is gradually gaining popularity. Rice, Jute, Sugarcane, Potato, Pulses, Wheat, Tea and Tobacco are the principal crops. The crop sub-sector dominates the agriculture sector contributing about 72% of total production. Fisheries, livestock and forestry sub-sectors are 10.33%, 10.11% and 7.33% respectively.

Bangladesh is the largest producer of Jute. Rice being the staple food, its production is of major importance. Rice production stood at 20.3 million tons in 1996-97 fiscal year. Crop diversification program, credit, extension and research, and input distribution policies pursued by the government are yielding positive results. The country is now on the threshold of attaining self-sufficiency in food grain production. Through the contribution of agriculture to the national economy and employment may diminish further, it will remain the single largest contributor to income and employment generation and a vital element in the country's challenge to achieve self-sufficiency in food production.

2 Agriculture or agricultural work

Agriculture means the activities relating to the production of vegetation and animal resources to meet the daily requirements of human beings. So agriculture includes such productive works which are related to the production of crops, rearing up of birds and animals, fisheries, a forestation etc. Agriculture is considered as the main profession in most of the developing countries of the modern world. In Bangladesh also agriculture is the main profession of its inhabitants.

3 Subsistence agricultural work and agricultural work for commercial production

The agricultural work can be directed with two motives; Such as subsistence agriculture and agricultural work for commercial purpose.

- **Agricultural work for subsistence:** From the agricultural work when one can mainly meet the self requirement, that agricultural work is known as subsistence agriculture. In the subsistence type of farming a nominal surplus remains after meeting the family expenditure by which farmer somehow continues agricultural work. The

following features can be observed in subsistence agricultural farming as mentioned below:

- 1 Subsistence farming is meant for the survival of the self and the members of the family only. So, this sort of agriculture is primarily administered for the consumption purpose of the family.
 - 2 In this form of agriculture the farmers engage only his family members, No hired laborer is engaged.
 - 3 In subsistence agriculture the owner himself takes part.
 - 4 Those agricultural products are produced which are consumed by the members of the family themselves.
- **Agricultural Activities for Commercial Purpose:** When agricultural production is undertaken and products are sold in the market with the Objective of earning profits is called agriculture for commercial purpose. Such agricultural activities are administered in such a way that profits can be earned through selling the markets. Agricultural activities for commercial purpose possess the following characteristics:
 - a In the commercial production the owner himself does not take part in agricultural activities; rather he engages laborer on hire for agricultural activities.
 - b In this sort of agriculture the main thrust is on profit, not consumption. For this efforts are made to make profit by selling whatever is produced. The family may not consume the produce. So in this type of farming family consumption does not get that much importance.

Basically for the sake of commercial purpose such agricultural activities are done. So those agricultural produce where prospect of profit is high are preferred.

4 The comparative discussion on subsistence agriculture and agriculture for commercial purpose

From the viewpoints of advantages and disadvantages the difference can be shown between self-supporting agriculture and agriculture for commercial production. These are discussed below:

- A. **Production:** Substance farmers produce only to meet the requirements of their family. If there is small surplus, he earns insignificant income from their sale. As a result his saving is very low; due to lack of requisite security he cannot produce loans. Under this situation he is not in a position to apply developed agriculture inputs in production. Consequently per acre yield becomes low.
- On the other hand, in case of commercial type of agriculture production, there exists

constant effort to produce for earning profits. In this case production is undertaken on a large-scale and farmer earns sufficient profit. When the income is high, savings will be high. These savings can be used by the farmers for purchasing better agriculture inputs and their use. Consequently, per acre production increases.

- B. **Risks:** In most cases production of agriculture produce upon nature. In case of subsistence agriculture the farmer produce only for his survival. In this circumstance if there is any national calamity and loss of produce he faces the crisis.

But when the agricultural works are done for commercial production the farmers produce one or two crops of substantial amount. As a result the loss incurred in case of one crop can be compensated by another. Besides he can save also. He can overcome the crisis if faced, with that savings, so from the view point of risk bearing it can be said that agriculture production on commercial basis is far better than subsistence agriculture.

- C. **Specialization:** If a person does the same work again and again, the efficiency which is acquired is called specialization. In the subsistence agriculture where only the family requirements are met, a number of crops are produced. In this case specialization cannot be attained. For this reason in case of substance agriculture advantages of specialization such as less cost, standard quality goods etc. are not available.

On the other hand since agriculture on commercial basis is directed towards the primary objectives of earning profit, the farmers concentrate on the production of one or two crops. As a result the cost of production is less, the quality improves and volume of production increase.

- D. **Marketing of products:** The process through which the product from the agriculture fields reaches to final consumers is called marketing, if marketing of is perfect or well organized, the farmers get fair price of their products. In subsistence farming hardly there is surplus production. As a result the farmer has no urge to think over the sale of products. At a low price he sells the surplus at his door-step. As a result he is deprived of fair price of his product.

But in agriculture for commercial production the farmer produces larger quantity of any crop. He always thins about making maximum profit at a minimum cost. As a result he earns profit through proper marketing of his product.

From the above discussion it can be said that for conducting production in agriculture successfully with efficiency and certainly, agriculture on commercial basis is far better than substances farming.

In Bangladesh most of the farmers are engaged in substances farming. For this there is hardly any surplus in agriculture production. Consequently the income of the farmers does not increase. In order to ensure more investment to agriculture through creating surplus more emphasis should be given to agriculture on commercial production rather than the substance agriculture.

5 Opportunities and Constraints of Agriculture in Bangladesh

Opportunities:

1. Agricultural sector is the single largest contributor to GDP.
2. Crop production system is highly labor intensive and there is an abundance of labor in the country.
3. Agriculture is the largest source of employment for skilled and unskilled labor.
4. Favorable natural environment generally exists throughout the year for crop production.
5. Wide range of bio-diversity exists for different crops.
6. Different crops and agricultural commodities are the main sources of nutrition including protein, minerals and vitamins.
7. Agricultural commodities have comparatively higher value addition than non-agricultural commodities.

Constraints:

1. Agriculture is dependent on the vagaries of nature and is risky.
2. Availability of cultivable land is decreasing.
3. Widespread poverty among the population engaged in agriculture.
4. Lack of required capital for agricultural activities.
5. Inadequacy of appropriate technology considering farmers socio-economic conditions.
6. Uncertainty of fair price of agricultural commodities due to underdeveloped marketing system.
7. Agricultural commodities are rapidly perishable and post harvest losses are too high.
8. Limited knowledge of common people about the nutritional value of agricultural commodities including vegetables and fruits.

6 Issues and challenges of agriculture in Bangladesh

1. **High level of rural poverty:** Poverty in Bangladesh is primarily a rural phenomenon, with 53% of its rural population classified as poor, comprising about 85%

of the country's poor. Achieving the millennium development goal (MDG) of having poverty to 26.5% by 2015 will require a growth rate of at least 4.00% in agriculture and 7.00% in the non-farm sector. However economic and institutional realities, the country's geographical and demographic characteristics and its vulnerability to natural disasters, make this a very challenging task.

2. **Low agricultural productivity:** Another challenge is rapidly shrinking land base. While the country's population is growing at the rate of 1.6% per year, demographic pressure and increased urbanization have caused cultivated area to decline at the rate of 1% per year. As cropping intensity has approached its limit (about 175% now) growth will need to come from intensification of cereal production, diversification into high-value crop and non-crop activities, and value addition in the agro-processing sector, including storage, processing and marketing. This will require reforming the agricultural research and extension systems, and financial and other regulations. Land administration and security issues also need to be addressed.
3. **Poorly functioning input and output markets:** The lack of easily accessible Markets and the traders significant constraints in both agricultural input and Output markets. Increase productivity high relative to services provided. Lack of Market information and infrastructure, the poor law and order situation, the existence of syndicates, and collection of illegal tolls further aggravate the Situation
4. **Lack of enabling rural investment climate:** For nearly 45% of the rural population, who are landless or functionally landless (owing less than 0.05 acre of land) and a majority of the new labor force every year, a declining land base and a small urban employment means that employment in the rural non-farm sector, however is constrained by lack of or poor quality of rural infrastructure and services, highly centralized government framework, weak rural financial systems, and a poor law and order situation.
5. **Weak rural institutions:** while the NGO sector in Bangladesh is well developed and the quality of informal institutions is improving, formal rural institutions remain very weak. Government agencies at all levels face overlapping functions, lack of coordination, low skill levels and incentives, and lack of responsibility, exacerbated by an urban bias. Elite capture is also quite common in rural areas.
6. **Vulnerability to natural disasters:** Bangladesh is the terminal floodplain delta of three large rivers -Ganga, Brahmaputra and Megna. Every year about 20 to 30 percent, and every few years 40%, of the country is flooded, causing serious damage to infrastructure, crops and overall economy. Projected climate changes and rise in the sea level are likely to worsen the situation. Since independence in 1971, the government has made large investment to protect against floods and cyclones. However issues such as public and private roles and community participation in disaster management, environmental protection, and institutional reforms of Bangladesh Water Development Board (BWDB), need to be addressed.

7 Priorities areas for agricultural development

1. Increasing agricultural productivity, diversification and value addition:

- (a) **Agricultural research and extension:** Both public and private investments in a dynamic and responsive agricultural research and extension system is essential to accelerate the transition from subsistence to commercial farming through diversification, export promotion, and bridging yield gaps.
- (b) **Enabling rural business environment:** To stimulate RNF growth, the country needs to provide an enabling rural business environment by investing in rural infrastructure, reforming its rural finance mechanism, regulatory framework, land policy and administration, and public expenditures, creating an incentive for rural SMEs is agro-business improving the low and order situation and ensuring decentralized and accountable rural service delivery.

2. Improving factors markets access to assets and natural resources management:

- (a) **Agricultural land:** land is becoming a scarce commodity in Bangladesh and land grab(particularly of public land) by the powerful is quite common there is thus a need to review land administration, ownership distribution, rights and titles and land use policy followed by enforcement law and policies.
- (b) **Agricultural inputs:** Use of quality agricultural inputs such as seeds, fertilizer, pesticides, farm equipment, and irrigation is critical to promote diversification, and increase productivity, however, despite major improvements, agricultural inputs continue to experience problems such as lack of timely availability, poor quality, and high price, that are even more market in remote rural areas. There is therefore a need to examine the impact of input price policy and subsidies on profitability and competitiveness, and explore options for their reform.
- (c) **Rural finance:** Here is a need to understand the constraints to access to rural finance; particularly by the missing middle farmers as well as SMEs. There is also a need to reform agricultural banks, to improve their recovery rate, reduce defaults, and increase the numbers of commercial bank branches in the rural areas.
- (d) **Water Resources Management:** There is a need to institutionalize participatory water management through water management organization, improve operations and maintenance of flood control infrastructures, and strengthen water sectors institutions, particularly BWDB and the water resources planning origination (WARPO). There is also a need to promote information sharing to reduce downstream flood damage.
- (e) **Natural Resources management:** In view of the growing threat to the longer term sustainability of natural resources, there is a need to design and enforce a policy and institutional framework for natural resources management and user

including anticipation. This will be critical to sustaining high agricultural growth in a country like Bangladesh with poor natural resources and high population density.

3. Strengthening Rural Institutions and Livelihood Support:

- (a) **Rural service delivery:** Improving Physical and social infrastructure roads, electricity, communication, water and sanitation, health and education in rural areas is fundamental both for promotion employment opportunities and welfare. While Bangladesh has done well and developing rural roads, it has a long way to go to meet other infrastructure needs, such as electricity, which is only available to 15 percent of villages. Because of the inefficiency of the government in public service delivery, the nongovernmental sector has de facto become one of the main actors in development efforts. While decentralization efforts remain incipient, the efficiency and sustainability of public service delivery remains in question.
- (b) **Livelihood Support:** Creating and strengthening local organizations such as water management organizations farmer associations, womens groups and village development comities is also essential to improving service delivery and accountability. Empowering rural communities, especially women, to create livelihood opportunities, such as through micro-credit programs is particularly important in this regard. Increasing the capacity for collective action has effectively enhanced the communities bargaining power, their access to assets and confidence in micro enterprise development.

8 Importance of agriculture in Bangladesh

The importance of agriculture in the economic development of an agricultural country is very high. Being an agriculture country the economic development of Bangladesh is depends basically on the development of its agriculture, from different viewpoints the importance of agriculture is discussed below:

- 1. **Main occupation:** About 75% people of Bangladesh, depend directly or indirectly on agriculture for their livelihood. Such a large scale dependence of the people on agriculture testifies the importance of agriculture in the economy of our country.
- 2. **Supply of food:** The different products of agriculture such as food grains, fish meat, milk, egg etc. meet the demand for food of the people. If the production from agriculture increases, it will be possible to supply more food. At the same time the existing food deficit in the country will be removed.
- 3. **Supply of raw materials for industries:** The major portion of the supply of necessary raw materials required for our existing industries and new industries to be

built up basically comes from agriculture. On the basis of availability of the raw materials like jute, sugar, tea, tobacco etc. the jute sugar tea and tobacco industries have been established in our country.

4. **Increase the tendency to establish agriculture related industries:** With the development of agriculture the demands for inputs like chemical fertilizer, insecticides, deep and shallow tube wells, power tiller, tractor etc. increase. For this there is a tendency to establish these industries. So if there is development in agriculture the industries like chemical fertilizer pesticides, deep and shallow tube well, power tiller, tractor industries will be established. As a result there will be rapid industrialization in the economy.
5. **Supply of cloths:** Agriculture supplies clothes to the people of our country. Cloths are made of jute, cotton, wool of sheep etc. the major portion of the demand for cloth is met up from these materials. Besides higher quality of carpets, and screen cloths are made from jute.
6. **Supply of inputs for house making and supply of fuel:** In the rural area of Bangladesh most of the houses are mainly made of bamboos, straw, leaves of trees, cane, wood etc. these are supplied by agriculture. Moreover wood, dry leaves of trees, jute sticks, corn husks, cow dung etc meet our demand for fuel in a greater amount. So by supplying the inputs of house making agriculture helps the construction of houses in rural areas. At the same time, it also helps to solve the accurate shortage of foal by supplying the inputs of fuel.
7. **Establishment of medicine industry:** We get raw materials of life saving drugs from agriculture. Many kinds of medicinal herbs grow here and there near houses and bushes in Bangladesh. On the basis of these medicinal herbs many herbal medicine industries have been built. From ancient days our rural folks use herbal medicine for the treatment of various diseases
8. **Main source of national production:** The volume of our national product mainly depends on agriculture. About 40% of our national production comes from agriculture. So the importance of agriculture as a single source to the national production is important. The more the volume of agricultural produce more will be our national and per capita production.
9. **Helping the earning of foreign exchange:** A considerable amount of foreign exchange is earned from the exports of different agricultural products. So it will be possible to earn more foreign exchange, if agricultural products are exported. This foreign exchange may be used for importing essential raw materials, machinery, fuel etc. for industries. As a result the speed of industrialization will be faster.
10. **Increase public revenue:** A good amount of public revenue comes from agriculture sector. Government of Bangladesh earns a lot of money from agricultural income tax,

fare from transportation of agricultural commodities, export duties of agricultural products etc. the increased amount of revenue can be spent for administration of the country and for its development.

11. **Creation of markets for industrial products:** The income of the cultivators increases with the development of agriculture. So their demand for industrial product also increases. As a result existing industries increase their production. Again, new industries are established to satisfy the increased demand. So for industrial development of the country the development of agriculture is essential.
12. **Rapid economic development:** The contribution of agriculture as single source to national production is remarkable. Our internal resource will increase development. So we can meet the increased cost of different development plans by internal resources. So through the development of this sector economic development is possible.

From the above discussion it is clear that the importance of agriculture in the economic development of Bangladesh is beyond any limit. The economic development or the overall development of our country depends on the very development of agriculture.

9 Characteristics of Agriculture in Bangladesh

Bangladesh is an agricultural country. But in spite of that the condition of agriculture is not developed. All the characteristics of a backward agricultural system are noticed in our country. So agricultural productivity in our country is low in comparison with other developed countries. The main characteristics of our agriculture are discussed below:

1. **Undeveloped production method:** Most cultivators of our country until now follow age old and unscientific methods of cultivation. The majority of our cultivators are not even directly aware of improved methods and instruments used agriculture. So our agriculture is undeveloped and full defects.
2. **Low productivity:** Though Bangladesh is an agriculture country. Its productivity per acre is very low compared to others country. In Japan the production of rice per acre is more than 2 tons, where as in Bangladesh it is a bit greater than ton.
3. **Cultivation for subsistence:** The cultivators of our country basically do cultivation for their own subsistence only, so here the scope for producing huge surplus product which can be sold is very much limited. However, recently by using high yielding technology attempts are being to increase agricultural product.
4. **Subdivision and fragmentation of holdings:** The land holdings of the majority of cultivators of this are very small. Again the plots are scattered throughout the

field, it is impossible to cultivate these small and scattered plots under mechanized methods. This is a great obstacle on way of modernization of agriculture.

5. **Landless cultivators:** The majority of the cultivators of our country are landless. They cultivate other peoples land as share croppers or only as they laborers. As they are not owners of land, they dont feel interest in cultivation. So production from land is small.
6. **Uncultivated land:** Because of the lack of the suitable irrigation and drainage system a good part of our total cultivatable land has remain uncultivated till now. So in practice, production is less than the amount which could be possible produce.
7. **Dependence on rainfall:** The agriculture of Bangladesh is yet basically dependent on rainfall .if there is good and timely rainfall, crops grow well. If this is not so, the production of crops is hampered seriously. As a result national production decreases.
8. **The major portion of land is under the cultivation of food crops:** About eighty percent of our total cultivable land is used to produce food crops. So raw material necessary for industrial use are produced in smaller quantity.
9. **Defective ownership of land:** We find in equality in the ownership of land in cultivation. About thirty percent of the big and middle farmers own eighty percent of total land .where as seventy percent of small cultivators own only of land.
10. **Absentee owners of land:** Many persons living in town are the owners of good amount of cultivable land in country. Supervision work is almost absent because the owner of land lives far away from land. As a result, the amount of produce is less.
11. **Share cropping:** About twenty five percent of total cultivable land of this country is cultivated in system of share cropping. The production by the share cropper is less than what would be if land was cultivated by the owner himself. So in our country the productivity per acre is very low.
12. **Poor, unhealthy and illiterate peasants:** Due to lack of food, clothes, houses, medical facilities etc. the peasants of our country are unhealthy and non-energetic. As they lack general education they are orthodox, fatalists and believers in superstitions. Likewise they are ignorant of an indifferent to modern technology of cultivation as they lack education about agriculture.

On account of the above mentioned characteristics, Productivity in agriculture is very low. The income of the farmers is small and their standard of living is also very low. The recent trend of improvement in agriculture which is seen in the developing countries of the world is remarkably absent in Bangladesh.

10 Causes of the backwardness of agriculture in Bangladesh

Although Bangladesh is an agricultural country, its agriculture is very backward and full of defects. The causes of the backwardness of agricultural of this country are discussed below:

1. **Old Method of Cultivation:** Old method of cultivation yet prevails in Bangladesh. The use of modern scientific methods modern equipments and inputs are, so to say, almost absent. The result is that the amount of agricultural production does not increase up to satisfaction.
2. **Subdivision and Fragmentation of Holdings:** The land holdings of the most of the cultivators of our country are subdivided and these are scattered in the different places of the field. It is difficult to introduce modern technique of production in this kind of plots and supervise the works of production properly.
3. **Problem of Credit:** Most of the cultivators of our country are poor. So they require loans for cultivating land. But the loans are uncertain and risky. Because these are collected from non-government sources. The amount of loans given by government source is insufficient and the methods of sanctioning loans are complex and troublesome. So the cultivators cannot invest money according to the needs of cultivation. So the volume of production does not increase.
4. **Unequal Distribution of Land Ownership:** The ownership of land in Bangladesh is very much unequal and defective. About 30 percent people are the owners of about 80 percent land. More than half of the total cultivators are landless. This kind of unequal distribution of agricultural land is not helpful to increase production.
5. **Lack of Goods Seeds and Manure:** To increase the production of crops the use of high yielding varieties of seeds and chemical fertilizer are very important, but in our country these two factors are very scarce. Moreover, as the farmer is very poor, they cannot even purchase these things. As a result, the yield is low for the low quality of seed and lack of adequate fertilizers.
6. **Lack of Irrigation Facilities:** Regular supply of water is essential to continue the work of cultivation properly. But scientific irrigation works have not yet been introduced on large scale. So agriculture of this country basically depends on rainfall. But rainfall is irregular. Sometimes it is heavy and some other time lights. So cultivation is often disturbed.
7. **Natural calamities:** At most every year natural calamities like flood, cyclone, tidal bore etc. occurs in Bangladesh. As a result, there is huge damage of crops. So the cultivators get fewer amounts of crops than what they expect to have.
8. **Decrease in the Fertility of Land:** The plots of agricultural land in Bangladesh are used continuously for year after year and remain under the cultivation of same

variety of crops. This leads to shortage of some chemical elements in those plots of land. This shortage is not regained due to lack of fertilizer. So these plots of land are becoming barren day by day and producing less.

9. **Attack of pest and Insects and Crop Diseases:** At the attack of locus pest and insects and of crop diseases huge amounts of crops are damaged in the field every year. So the cultivators get fewer amounts of crops.
10. **Water stagnation and salinity of land:** On account of heavy rainfall and lack of good drainage system vast areas of land suffer from water stagnation. Moreover, due to salinity many areas at sea shores remain uncultivated. As a result, the amount of cultivable land has decreased.
11. **Weak rural infrastructure:** Infrastructure, such as, drainage system, roads supply of electricity, health centre, supply of drinking water etc, are very much important for agricultural development .but these are very insufficient in rural areas of Bangladesh. So, it is difficult to manage the process of production efficiently.
12. **Defective marketing system:** The marketing system of agricultural products is full of defects. Due to the presence of middlemen, acute insolvency of the cultivators, lack of god owns. Undeveloped transport and communication system etc. the peasants are bound to sell their products just after harvest at low prices. As a result, they dont get fair price of their product.
13. **Poverty and illiteracy of the peasant:** the majority of peasants of Bangladesh are poor deprived of the light of education. So they are orthodox fatalists and superstitious. They are ignorant and careless about the application of modern techniques in the field of production. Again in many cases they are against new techniques of production.
14. **Lack of subsidiary occupation:** Due to lack of subsidiary occupations in rural areas, the peasants cannot increase their income. As a result of this they cannot improve financial position and cannot be achieve so much in the field of production.
15. **Absence of proper organization:** the farmers in Bangladesh carry on agricultural activities in the small and scattered holdings in an unplanned simple way of management. There is lack coordination in the different stages of cultivation and among the input used. So the cultivation cannot conduct and manage agricultural works properly.
16. **Absence of well coordinated agricultural policy:** a well coordinated agricultural policy is necessary at the government level for continuous and rapid development of agricultural sector. Though Bangladesh is agrarian country, no long term well- coordinated policy has yet been undertaken. As a result, a well- coordinated technology is not being followed in our country. For reason, desired structural transformation is not taking place.

Because of these problems, in spite of being an agricultural country, Bangladesh is backward in the field of agriculture to a greater extent. So productivity per acre is also very low.

11 Modern technology in agriculture

The agricultural system of Bangladesh is very much underdeveloped and defective. So the agricultural productivity is very low and the income of the cultivator is very low. To remove this condition production should be run in such a way that the yield of crops increases, the income of the cultivator's increases and more investment in agriculture is possible. The different aspects of use of modern technology in agriculture are discussed below:

1. **Use of high yielding varieties of seeds:** To increase production in agriculture the use of high yielding varieties of seeds is very important. To remove food deficit in our country high yielding varieties of seeds of rice and wheat are being used. Bangladesh Rice Research Institute (BRRI) has invented different varieties of high yielding seeds of rice since 1967. The major achievements of the Bangladesh Rice Research Institute (BRRI) has been the development of 37 high yielding modern variety (MV) rice strains adaptable to different ecosystems since its establishment. Thirteen of these MV rice are suitable for cultivation in both the Boro and Aus seasons, seven in the Boro season five in the Aus season, and twelve in the transplanted Aman (T Aman) season. With appropriate management, and under favorable soil conditions, these MV rice may yield 5-6 m tons/ha in the Boro, 3-4 m tons/hectar in the Aus, and 4-5 m tons/ha in the T Aman seasons. In the production of wheat also the role of high yielding varieties of wheat seed is unique. During the period before independence the production of wheat per was 1 lac ton, whereas at present the production rate of wheat per year has been increased to 9 lac tons (2009-2010).
2. **Use of fertilizer:** Bangladesh is basically an agricultural country having only a limited land for agricultural purpose, which is also on the decrease because of increasing trend of industrialization, development of housing complexes for large population and increased salinity effect across the coastal zone. Whereas the demand of food-grains and other agricultural products have increased manifold with the population boom. The government has been emphasizing the agricultural sector for increasing the food grains production to meet up our huge demand. Our land is losing its fertility because of continuous harvesting for a long period for which use of makeup fertilizer is very much essential for increasing the food grain production. Fertilizer products cover urea, nitrogenous, potash, and phosphate fertilizers (including ground rock phosphate). Thus use of fertilizer will lead us self-sufficiency through green revolution for production of different agricultural products.
3. **Use of pesticides:** Bangladesh is predominantly an agricultural country with an area of 147 570 sq. km. Agriculture plays an important role in the lives of Bangladeshi

people. The major crops grown in the country are rice, wheat, jute, potato, sugarcane, vegetables and tea. The warm and humid climatic conditions of the country, increased use of modern high yielding varieties of crops and more use of chemical fertilizers are highly favorable for development and multiplication of pests and diseases. The estimated loss in yields due to attacks from pests and diseases annually ranges from 15 to 25 percent. Pest control in Bangladesh is dependent on the use of pesticides. The use of pesticides in Bangladesh is based on minimum curative applications with 70 percent in rice and the remaining 30 percent on tea, sugarcane, potato, mango, banana & vegetables.

4. **Irrigation:** Irrigation is an artificial application of water to the land or soil. It is used to assist in the growing of agricultural crops, maintenance of landscapes, and revegetation of disturbed soils in dry areas and during periods of inadequate rainfall. For continuous agricultural production it is necessary to irrigate the land according to needs. Especially in cultivation of high yielding varieties of seeds sufficient irrigation is needed. In our country agriculture basically depends on rainfall. But rainfall is irregular. Sometimes it is heavy and sometimes inadequate. So, cultivation of crops hampered. To remove these situations mechanized irrigation is necessary. At present irrigation works are done by means of deep and shallow tub wells, power pumps etc. in addition to these, irrigation works is done by cutting canals from rivers, hours of lifting water from rivers, canals etc. with the help of local appliances made of wood.

12 Strengthening Bangladesh's agricultural technology system to reduce rural poverty

Agriculture accounts for about 23% of Bangladesh's GDP and another 33% of GDP is contributed by the rural non-farm economy, which is largely linked to agriculture. Agriculture and rural non-farm economy are the main sources of livelihood for rural people. With almost 85 out of 100 poor people living in rural areas, poverty in Bangladesh is largely a rural phenomenon. Improved economic performance of the agriculture and the rural non-farm sectors is, therefore, critical to reducing poverty.

Overall, the agricultural sector has performed reasonably well. However, the growth in the agricultural sector has stalled in the last few years. In the Poverty Reduction Strategy Paper (PRSP) document, the Government of Bangladesh (GOB) has indicated the need to achieve 7% overall annual economic growth in order to achieve the Millennium Development Goal (MDG) of poverty reduction a target which will not be possible unless the agricultural sector achieves at least 5% annual growth rate. Among the key constraints in achieving this objective are:

- Agricultural land is actually declining and quality is deteriorating.
- Agricultural productivity growth has either stagnated or declined.

- Pace of diversification from low value cereals to high value commodities is slow.

One of the most important ways to address these constraints is to generate, disseminate and promote the use of appropriate agricultural technology that is suitable for the agro-climatic conditions in Bangladesh. This would require an efficient and responsive agricultural research and extension system. At the request of the GOB, the World Bank has agreed to assist with the strengthening of the agricultural technology system. A roadmap for improving agricultural productivity has been developed in a draft report on Revitalizing Agricultural Technology System in Bangladesh.

The proposed project will address the following issues:

1. **Promoting institutional efficiency of agricultural research system:** Enhancing the efficiency and effectiveness of the National Agricultural Research Systems (NARS) is vital to increase agricultural productivity. It is also essential to give Bangladesh's farmers a greater say in directing NARS, along with improving understanding of yield gaps (i.e., what is being produced and what is possible) and developing technologies to promote productivity, sustainability and diversification of agriculture.
2. **Improving sustainability of funding for agricultural research:** One major constraint to higher agricultural productivity is the low level of government expenditure on agricultural research. At present, the expenditure is only about 0.2% of Agricultural GDP, compared to 0.6% for other developing countries and over 2% for developed countries. In addition, the existing funding for research is unstable and often tied with the life span of a particular development project (i.e., funding stops when project ends). The project proposes to develop a system whereby funding for research can be sustained and allocated to priority research projects on a competitive basis.
3. **Decentralizing the agricultural extension system:** An agricultural extension system based on the bottom-up approach will be more responsive to farmers' needs and is necessary to improve the overall performance of the agricultural sector. Such a system would be more efficient, knowledge-based and address the real problems faced by farmers.
4. **Developing value chain and market linkages:** An element of strengthening agricultural technology systems would be to promote the development of value chains of a few selected high value commodities based on comparative advantage, farmer preference and market demand.

Finally, the World Bank has requested the status of GOB's strategy for safeguarding against the risk of avian influenza or "bird" flu and a possible role of the proposed project in preparing the agricultural sector to meet any such threat. Bangladesh has been identified

as a high risk country. All the appropriate surveillance, diagnostic and preventive activities must be in place to safeguard the country against such a risk.